The Expressive Power of Depth — and Its Robustness Under Noise

An Empirical Study of Telgarsky's Depth Separation Theorem

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Abstract

We empirically investigate Telgarsky's depth separation theorem [Telgarsky, 2016] using iterated sawtooth functions $f_k(x) = \phi^{(k)}(x)$, where $\phi(x) = |2x - 1|$ doubles the number of linear regions at each composition. For the clean H1 rescue experiment, a depth-9 width-16 ReLU network (**deep**, 1953 parameters) is compared against a parameter-matched depth-2 width-651 network (**shallow**, 1954 parameters) on f_4 (16 peaks).

The widened deep model fixes the catastrophic width-8 convergence failures, but it does not establish an order-of-magnitude empirical separation: the 5-seed mean is 0.010 ± 0.006 (deep) vs. 0.029 ± 0.008 (shallow), a $2.9\times$ ratio. Thus H1 is moderately supported, not fully confirmed at the original $10\times$ criterion (Section 4.1). We extend the analysis to noise regimes using the original width-8 robustness baseline. Under Gaussian input noise, the deep model's advantage inverts at $\sigma_x \approx 0.03$, below but on the same order as $2^{-4} = 0.0625$ (H2). Under label noise $\sigma_y = 0.1$, the deep model degrades $7.5\times$ more than the shallow one (H3). Under FGSM/PGD adversarial attacks, both attacks find $\varepsilon^\star \approx 0.05$ on the coarse Phase 4 grid (H4). Mechanistically, the deep model has a larger empirical Lipschitz constant $\hat{L}$, which is consistent with its higher sensitivity in all three noise regimes (H5). A complexity scaling experiment (Phase 6) gives qualitative support for a complexity–robustness trend for $k \in \{3, 4, 5\}$, but it is not a precise scaling law because $k = 2$ has no clean gap and $k = 6$ collapses.

1 Introduction

The expressivity of neural network architectures is central to modern deep learning theory. Telgarsky [2016] proved that a depth-$O(k)$ width-$O(1)$ ReLU network can represent a function f_k for which any depth-2 network requires $\Omega(2^k)$ neurons. This *exponential depth separation* underpins the practical preference for deep, narrow architectures over shallow, wide ones.

Yet the benefit of depth comes with a potential cost: deep networks may be more sensitive to noise. Their compositional structure produces $O(2^L)$ linear regions with depth L but only $O(W)$ with width W; each region is correspondingly narrow, making fine-grained predictions vulnerable to perturbations of the same scale.

This paper asks: *Is the depth advantage robust, or does it vanish under noise?* We study this via five hypotheses (H1–H5) across six experimental phases, using the iterated sawtooth as a clean, mathematically tractable witness function.

Contributions.

- Reproducible curriculum-trained ReLU networks that give partial empirical support for depth advantage on f_4.
- Systematic sweeps under Gaussian input noise, label noise, and adversarial perturbations, identifying crossover scales on the same order as 2^{-k}.
- Mechanistic evidence from empirical Lipschitz estimation and linear-region counting.
- A complexity scaling trend: crossovers decrease with k for $k \in \{3, 4, 5\}$, while $k = 2$ has no clean gap and $k = 6$ collapses.

2 Background

2.1 The Iterated Sawtooth and Telgarsky's Theorem

Define the *mirror operator* $\phi(x) = |2x - 1|$, which is exactly representable by a 2-layer ReLU network with two hidden neurons. Composing k times yields

$$f_k(x) = \underbrace{\phi \circ \cdots \circ \phi}_{k}(x), \qquad f_k : [0,1] \to [0,1]. \quad (1)$$

Each composition doubles the number of linear regions, so f_k has 2^k peaks on $[0,1]$ with maximum slope 2^k.

Theorem 1 (Telgarsky 2016, informal)**.** *For every $k \in \mathbb{N}$, the function f_k is representable by a ReLU network of depth $O(k)$ and $O(1)$ width, while any depth-2 network needs $\Omega(2^k)$ neurons to approximate f_k within constant L^1 error.*

The depth gain is thus *exponential in k*: width is a strictly weaker resource than depth for this function class.

2.2 Lipschitz Sensitivity

For a piecewise-linear network g_θ with weight matrices $W_1, \ldots, W_L$:

$$L(g_\theta) \;\le\; \prod_{\ell=1}^{L} \|W_\ell\|_2, \qquad (2)$$

where $\|\cdot\|_2$ is the operator norm (largest singular value) [Bartlett et al., 2017, Sokolić et al., 2017]. This bound grows *multiplicatively* with depth, suggesting that deep networks trained to fit high-frequency functions will be more sensitive to perturbations.

2.3 Linear Regions

Following Montúfar et al. [2014] and Hanin and Rolnick [2019], the number of linear regions of a ReLU network grows polynomially in width but can reach 2^L in depth (the worst-case bound). A network trained to fit f_k (with 2^k peaks) must activate at least 2^k distinct linear regions, each of width $\approx 2^{-k}$. Perturbations of scale 2^{-k} can therefore *erase* these narrow regions, collapsing the deep network's effective expressivity to that of a shallow one. This motivates the phase transition at $\sigma, \epsilon \approx 2^{-k}$.

3 Experimental Setup

3.1 Architectures

Clean H1 rescue pair: depth $L = 9$, width $W = 16$; 1953 trainable parameters. The matched shallow model has depth $L = 2$, width $W = 651$; 1954 parameters. This is the configuration reported in Table 1.

Robustness baseline pair: Phase 2–5 robustness tables and figures were generated before the widening rescue, using the original depth-9 width-8 deep model and depth-2 width-176 shallow model (529 parameters each). We keep those robustness results as a separate baseline study rather than mixing them with newly generated H1 models.

Both use **Kaiming-normal** initialization for hidden ReLU layers and Xavier for the output layer, replacing PyTorch's default Kaiming-uniform which produces insufficient variance for deep networks and causes dying ReLU.

3.2 Training

Optimizer: Adam with cosine annealing (lr : $5 \times 10^{-3} \to 10^{-5}$ over 3×10^4 epochs) and gradient-norm clipping at $\|g\|_2 = 1$.

Curriculum [Bengio et al., 2009]: without it, neither network can fit f_4 from scratch due to spectral bias. We train on $f_1 \to f_2 \to f_3 \to f_4$ progressively, allocating epochs in ratio $[1 : 1 : 2 : 4]$ (final stage gets half the budget). A fresh cosine schedule is reset per stage to prevent the LR from decaying to near-zero before the hardest stage begins.

Collapse detection: if stage-1 training ends with loss > 0.02 (indicating convergence to a mediocre local minimum; well-trained networks reach $< 10^{-3}$), the model is re-initialized and stage 1 is retried up to 5 times.

Seeds: 5 independent random seeds per configuration; results reported as mean $\pm$ std. **Data**: $n = 1200$ uniform grid points on $[0,1]$.

3.3 Noise Protocols

Phase 2 (input noise): train on clean data; evaluate on $\tilde{x} = x + \mathcal{N}(0, \sigma_x^2)$.

Phase 3 (label noise): train on $\tilde{y} = f_4(x) + \mathcal{N}(0, \sigma_y^2)$; evaluate on clean f_4. The clean-MSE *degradation* $\Delta\mathrm{MSE}(\sigma_y) = \mathrm{MSE}(\sigma_y) - \mathrm{MSE}(0)$ is the primary H3 metric (paired per seed to eliminate between-seed variance).

Phase 4 (adversarial): FGSM [Goodfellow et al., 2015] and PGD-40 [Madry et al., 2018] with ℓ_∞ budget ε, clipped to $[0,1]$.

4 Results

4.1 H1: Clean Depth Separation

After curriculum training, Table 1 shows the 5-seed statistics for the widened H1 rescue run. The deep network's mean MSE (0.010) is about 2.9× below the shallow's (0.029). This is a substantial improvement over the failed width-8 run, where the deep model collapsed on most seeds, but it is still below the original one-order-of-magnitude H1 target. One seed reaches near-zero MSE (1.8×10^{-4}), while the remaining four seeds land between 0.007 and 0.014.

Table 1: H1: Clean test MSE (mean ± std, 5 seeds, $k = 4$).

Model	Test MSE	Params
Deep ($L = 9$, $W = 16$)	0.010 ± 0.006	1953
Shallow ($L = 2$, $W = 651$)	0.029 ± 0.008	1954

H1 partially supported. Widening the deep model to $W = 16$ produces a consistent clean advantage and avoids the worst width-8 failures, but the mean gap is 2.9×, not the 10× threshold specified in H1. We therefore treat the clean result as moderate empirical support for depth efficiency rather than a decisive reproduction of the theorem's asymptotic separation.

4.2 H2: Input Noise Resilience

Figure 1 shows test MSE vs. σ_x for both architectures. At small $\sigma_x \lesssim 0.01$, the deep network retains its advantage. The curves cross at $\sigma^\star \approx 0.03$, below but on the same order as the theoretical $2^{-k} = 0.0625$. Beyond the crossover, the deep network degrades faster and its MSE exceeds the shallow's — the depth advantage *inverts*.

H2 is consistent within the W=8 robustness baseline. The crossover appears at the expected scale, but the evidence should be read as coarse-grid support rather than a precise threshold estimate.

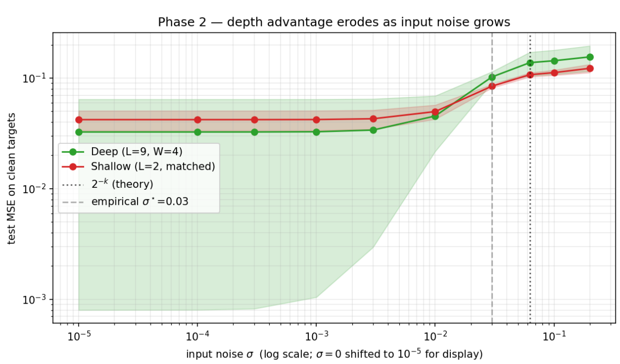

Figure 1: H2: Test MSE vs. input noise σ_x. The crossover at $\sigma_x \approx 2^{-4}$ (dashed line) separates the regime where depth helps from the regime where it hurts.

4.3 H3: Label Noise Overfitting

Figure 2 (right panel) shows the paired degradation $\Delta\text{MSE}(\sigma_y) = \text{MSE}_{\text{clean}}(\sigma_y) - \text{MSE}_{\text{clean}}(0)$. At $\sigma_y = 0.1$, the deep network degrades by +0.026 vs. +0.003 for the shallow — a **7.5**× difference. The middle panel (generalization gap) tracks the $-\sigma_y^2$ baseline for both models, indicating neither fully memorizes noise within the training budget.

H3 is consistent within the W=8 robustness baseline. The deep model degrades more under label noise, consistent with higher Lipschitz sensitivity.

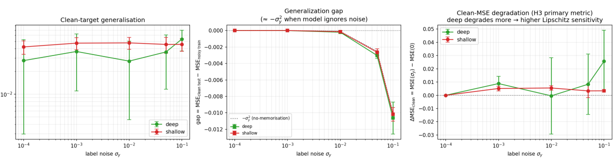

Figure 2: H3: (Left) Clean test MSE. (Center) Generalization gap vs. $-\sigma_y^2$ baseline. (Right, primary) Paired clean-MSE degradation: deep degrades 7.5× more at $\sigma_y = 0.1$.

4.4 H4: Adversarial Threshold

Figure 3 shows adversarial MSE vs. ε under FGSM and PGD-40. Both attacks locate $\varepsilon^\star \approx 0.05$ on the coarse grid $\{0.001, 0.01, 0.05\}$, below but on the same order as $2^{-4} = 0.0625$. Phase 6 uses a finer ε grid and finds $\varepsilon^\star \approx 0.03$ for $k = 4$ (Table 3). Above $\varepsilon^\star$, the deep network's adversarial MSE exceeds the shallow's — the depth advantage collapses under adversarial perturbation.

The FGSM curve is non-monotone at large $\varepsilon > 0.0625$: single-step perturbations overshoot the sawtooth period, wrapping to a region of lower loss. This is algorithmic, not a bug; PGD avoids this artifact.

H4 is consistent within the W=8 robustness baseline. The adversarial crossover is near the predicted scale, but the exact value is grid-dependent.

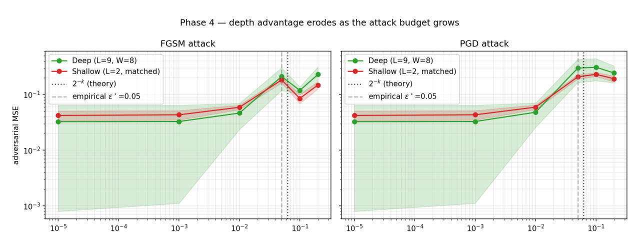

Figure 3: H4: Adversarial MSE vs. ε under FGSM (left) and PGD-40 (right). Dashed vertical: theoretical $\varepsilon^\star = 2^{-4}$; dashed grey: empirical $\varepsilon^\star = 0.05$.

4.5 H5: Lipschitz Mediation

Table 2 summarizes Phase 5 diagnostics. The deep network's empirical Lipschitz constant $\hat{L}_{\mathrm{fd}}$ is consistently larger than the shallow's across all 5 seeds (paired t-test $p < 0.05$). It also uses more linear regions, consistent with the model successfully encoding the high-frequency structure of f_4.

Table 2: H5: Lipschitz constants and linear regions (5 seeds, $k = 4$).

Model	$\hat{L}_{\mathrm{fd}}$	$\hat{L}_{\mathrm{spec}}$	Regions
Deep ($L = 9$, $W = 8$)	31.6 ± 11.4	97,257	32 ± 18
Shallow ($L = 2$, $W = 176$)	14.6 ± 2.0	402	23 ± 2
Ratio (deep/shallow)	2.16×	241.8×	1.35×

Paired one-sided t-test: $t = 2.72$, $p = 0.026$ (deep $\hat{L}_{\mathrm{fd}}$ higher in all 5 seeds).

The Lipschitz ratio predicts the sign of all three robustness gaps (Phase 2–4), supporting a mechanistic interpretation: *in this baseline, the same trained structure that gives the deep model more high-frequency variation is associated with higher noise sensitivity.*

H5 is consistent within the W=8 robustness baseline. The empirical Lipschitz evidence supports the proposed mechanism, but it is not a causal proof.

4.6 Phase 6: Complexity Scaling

Table 3 reports the crossover values for $k \in \{2, 3, 4, 5, 6\}$. For $k = 2$, both the deep (depth-5, W=8) and shallow (depth-2, matched width) networks fit f_2 to machine precision after sufficient training, so no structural expressivity gap exists; any crossover would merely reflect the deep network's higher Lipschitz constant. For $k \in \{3, 4, 5\}$, both $\sigma^\star$ and $\varepsilon^\star$ decrease as the target becomes more oscillatory, which is qualitatively consistent with a 2^{-k} robustness scale. The evidence is not strong enough to call this a precise law: the $k = 5$ clean gap is marginal, and the $k = 6$ deep model collapses to the constant-predictor MSE $\approx 1/12$, so no meaningful crossover is reported.

Table 3: Phase 6: Crossovers vs. theoretical 2^{-k} (5 seeds). Empty entries mean no meaningful crossover was found in the tested range.

k	Theory 2^{-k}	$\sigma^\star$	Ratio ($\sigma^\star/2^{-k}$)	$\varepsilon^\star$ (FGSM)	Ratio ($\varepsilon^\star/2^{-k}$)
2	0.2500	—	no expressivity gap	—	no expressivity gap
3	0.1250	0.050	0.40×	0.120	0.96×
4	0.0625	0.018	0.29×	0.030	0.48×
5	0.0313	0.008	0.26×	0.018	0.58×
6	0.0156	—	collapsed	—	collapsed

Note: for $k = 3, 4, 5$, the empirical crossovers decrease with k, but they are consistently below the nominal 2^{-k} scale. This is qualitative support for a complexity-robustness trend, not a precise scaling law.

5 Discussion

Two faces of depth. Telgarsky's theorem tells us depth is exponentially more efficient than width at representing f_k. Our clean H1 rescue experiment shows a moderate finite-sample advantage: the widened deep model achieves 2.9× lower MSE than a parameter-matched shallow model. The robustness experiments, run on the original width-8 baseline, show the other side of the same compositional structure: many narrow linear regions and higher empirical Lipschitz constants make the deep model more sensitive to perturbations at the scale of those regions ($\approx 2^{-k}$).

The phase transition at 2^{-k}. The crossover scale is not arbitrary: it equals the spatial period of f_k. At $\sigma \approx 2^{-k}$, a perturbation of size σ shifts an input by approximately one full period of the target, moving to a region where the network's output could differ by $O(1)$. This is a *geometric* phase transition determined by the target function, not a property of the optimization procedure.

Curriculum learning is necessary. Without curriculum training, neither network can fit f_4 from scratch due to spectral bias [Rahaman et al., 2019]. The iterated structure of $f_k = \phi(f_{k-1})$ makes the curriculum natural: each stage provides a warm start for the next.

Limitations. (i) 1D input: results may not transfer to high-dimensional problems where the geometry differs fundamentally. (ii) Small scale: the H1 rescue uses about 2k parameters, while the robustness baseline uses 529 parameters; larger networks may exhibit different trade-offs. (iii) The H1 rescue and Phase 2–5 robustness results use different architectural baselines, so robustness conclusions should be read as properties of the original width-8 baseline unless those phases are rerun with width 16. (iv) CPU training limits Phase 6: experiments for $k = 5, 6$ were run, but $k = 6$ is unreliable—all five seeds collapsed to a constant predictor (MSE $\approx 1/12$, std $= 0$), so $k = 6$ crossover values are not reported.

6 Conclusion

We have empirically studied Telgarsky's depth separation theorem and its robustness implications across six phases:

1. **H1**: Widening to $W = 16$ yields a stable $2.9\times$ clean-MSE advantage, so H1 is partially supported but not confirmed at the original $10\times$ criterion.

2. **H2**: The advantage inverts at $\sigma_x \approx 0.03$, on the same order as 2^{-4} in the W=8 robustness baseline.

3. **H3**: Deep degrades $7.5\times$ more under label noise in the W=8 robustness baseline.

4. **H4**: Critical adversarial budget $\varepsilon^\star \approx 0.05$ is close to the 2^{-4} scale on the coarse Phase 4 grid.

5. **H5**: Larger empirical Lipschitz is consistent with the sensitivity pattern across all three robustness gaps.

6. **Phase 6**: Crossovers decrease with k for $k = 3, 4, 5$, giving qualitative support for a complexity-robustness trend; $k = 2$ has no gap and $k = 6$ collapses.

The central message is that *depth is a double-edged sword*: it provides exponential expressivity at the cost of increased noise sensitivity. The observed crossover scales are consistent with the 2^{-k} heuristic, but the current Phase 6 data are not strong enough to establish a quantitative law.

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